# CyberTicket

## A Case Study in SOC Automation and Efficiency

*Implementation Report and Outcomes Analysis*

https://cyberticket.netlify.app/

## Executive Summary

CyberTicket is a browser-based security incident ticketing tool developed to streamline the process of creating standardized M365 security tickets from Wazuh SIEM logs. This report outlines the rationale behind its development, the iterative improvement process, and the measurable outcomes achieved through its implementation in the Security Operations Centre (SOC).

## 1. Project Rationale: Why CyberTicket Was Developed

### The Problem Statement

Prior to CyberTicket's development, SOC analysts faced significant challenges when processing M365 security alerts:

**Manual, Time-Intensive Processes:** Analysts manually extracted relevant fields from logs in the SIEM, including timestamps, user accounts, IP addresses, operations, and user-agent strings. This process was repetitive and prone to human error, particularly during high-alert periods.

**Inconsistent Ticket Quality:** Without a standardized template, ticket formatting varied between analysts. Critical fields were occasionally omitted, and the level of detail provided was inconsistent, impacting downstream investigation efficiency.

**Lack of Threat Intelligence Integration:** IP reputation data from AbuseIPDB required separate manual lookups and data entry, adding 2–5 minutes per ticket and creating workflow fragmentation.

**Cognitive Load:** Analysts spent valuable mental resources on administrative ticket creation rather than analytical work, leading to alert fatigue and reduced job satisfaction.

**No Metrics Capture:** Time-to-detect calculations were performed manually and inconsistently, making it difficult for the SOC to provide TTD figures accurately.

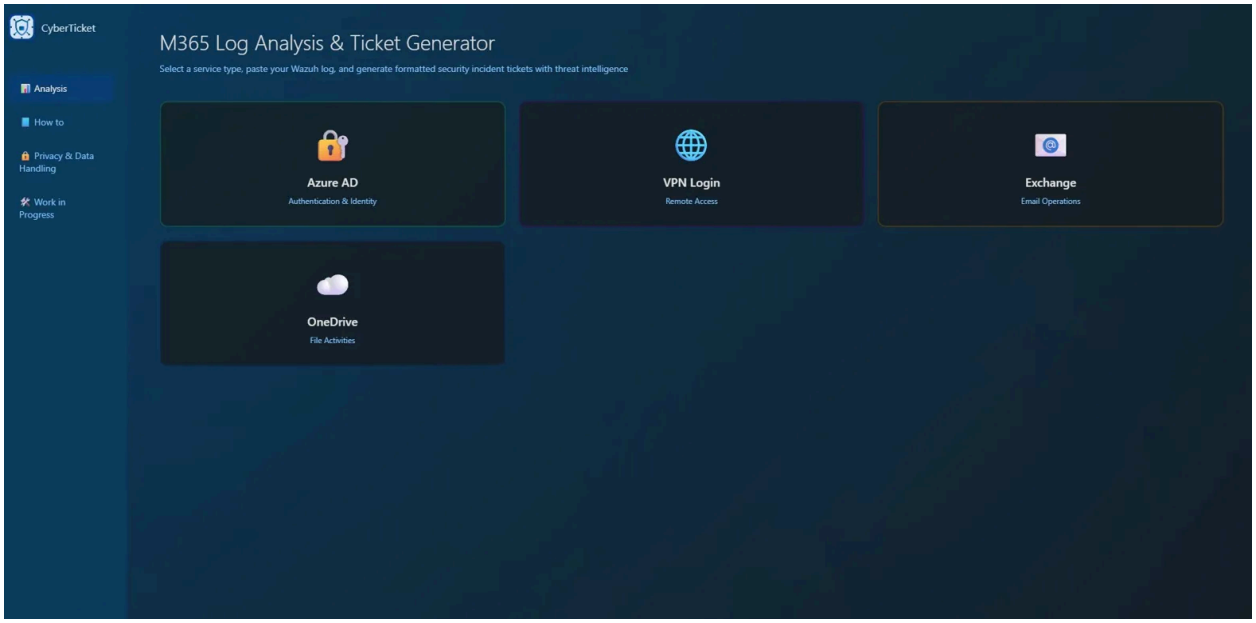


*Figure 1: CyberTicket landing page showing service selection interface (Azure AD, VPN Login, Exchange, OneDrive)*

## Development Objectives

CyberTicket was created to address these pain points through:

- Automation of data extraction from SIEM JSON logs
- Standardization of ticket formatting across all analysts
- Integration of real-time threat intelligence from AbuseIPDB
- Reduction of time spent on administrative tasks
- Enabling reliable performance metrics through time-to-detect calculations
- Privacy-by-design architecture with client-side processing only

# 2. The Iterative Improvement Process

## Initial Development Phase

The first iteration of CyberTicket focused on basic functionality:

- Azure AD (AAD) log parsing
- Simple field extraction (timestamp, user, IP, location)
- Basic HTML output formatting

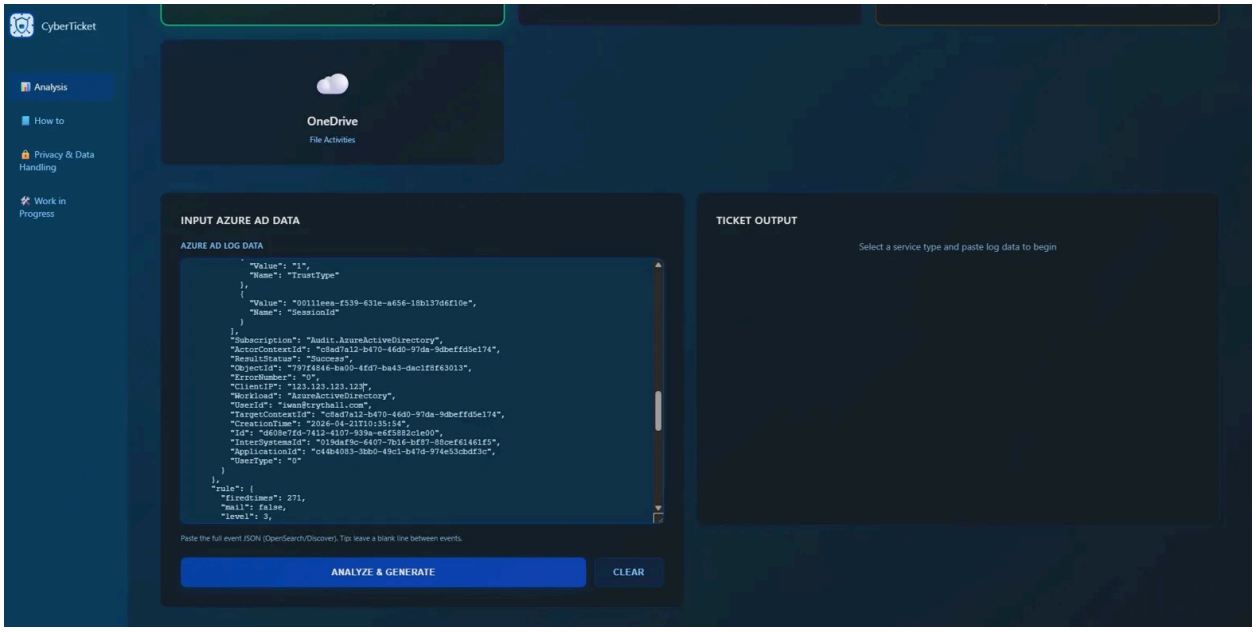


*Figure 2: Input interface showing Wazuh event JSON being pasted for analysis*

## Continuous Enhancement Cycle

CyberTicket has undergone ongoing refinement based on analyst feedback and operational requirements:

**Phase 1 — Service Expansion:** Added support for Exchange and OneDrive logs, recognizing that M365 security encompasses multiple services requiring different field sets and formatting.

**Phase 2 — Threat Intelligence Integration:** Implemented AbuseIPDB API integration via Netlify serverless functions to provide:

- Abuse confidence scores
- ISP and usage type information
- Historical abuse reports
- Country/City of origin data

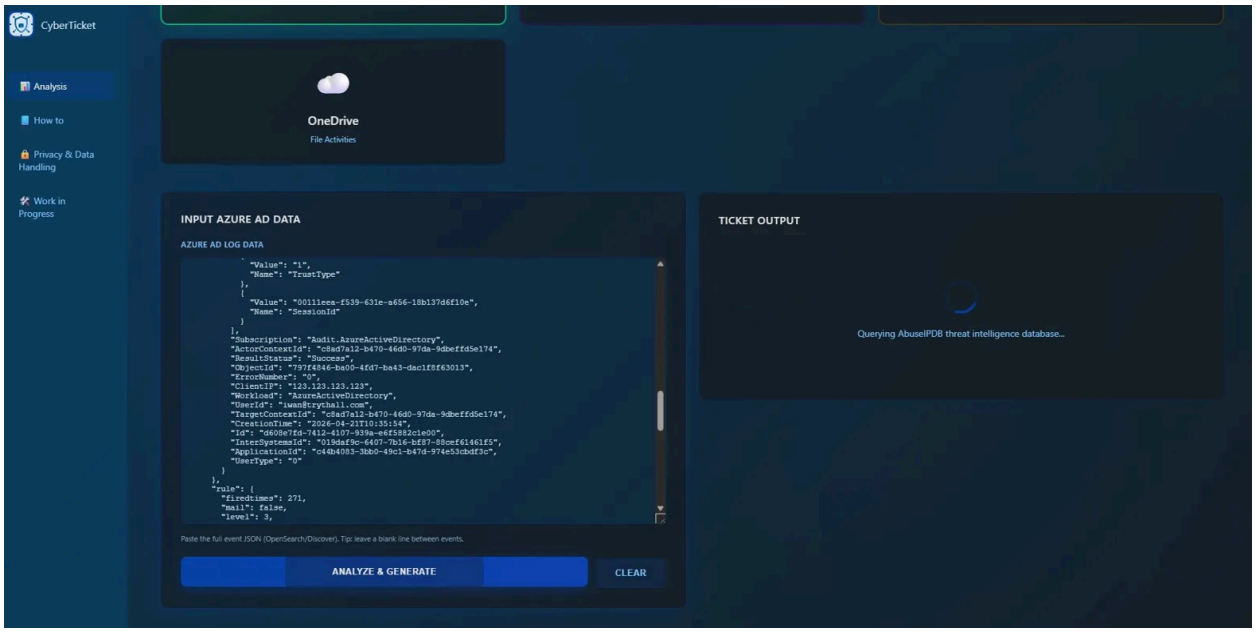


*Figure 3: Real-time threat intelligence lookup in progress, querying AbuseIPDB database*

**Phase 3 — Multi-Log Support:** Enabled processing of multiple events in a single ticket with automatic numbering, addressing the common scenario where related malicious activities generate multiple SIEM alerts.

**Phase 4 — Time-to-Detect Calculator:** Added a dedicated feature to calculate the time between log generation and detection, providing quantifiable metrics for SOC performance monitoring.

**Phase 5 — Subject Line Automation:** Implemented dynamic subject line generation following the standardized format: SOC Ticket Pre-Notification: M365, CLIENT-Shorthand, [Activity Description], P4, ensuring consistency in ticketing system notifications.

**Phase 6 — User Experience Refinement:** Streamlined the interface by removing redundant buttons, optimizing button placement, and improving the copy-paste workflow for faster ticket submission.

## Development Methodology

The improvement process followed a feedback-driven approach:

1. **Observation** of analysts using the tool during daily operations
2. **Collection** of pain points and feature requests
3. **Prioritization** based on frequency of use and impact
4. **Rapid iteration** with immediate deployment to production
5. **Validation** through analyst usage and satisfaction monitoring

This agile methodology enabled quick adaptation to changing SOC needs without lengthy development cycles or vendor dependencies.

## 3. Outcomes and Measurable Impact

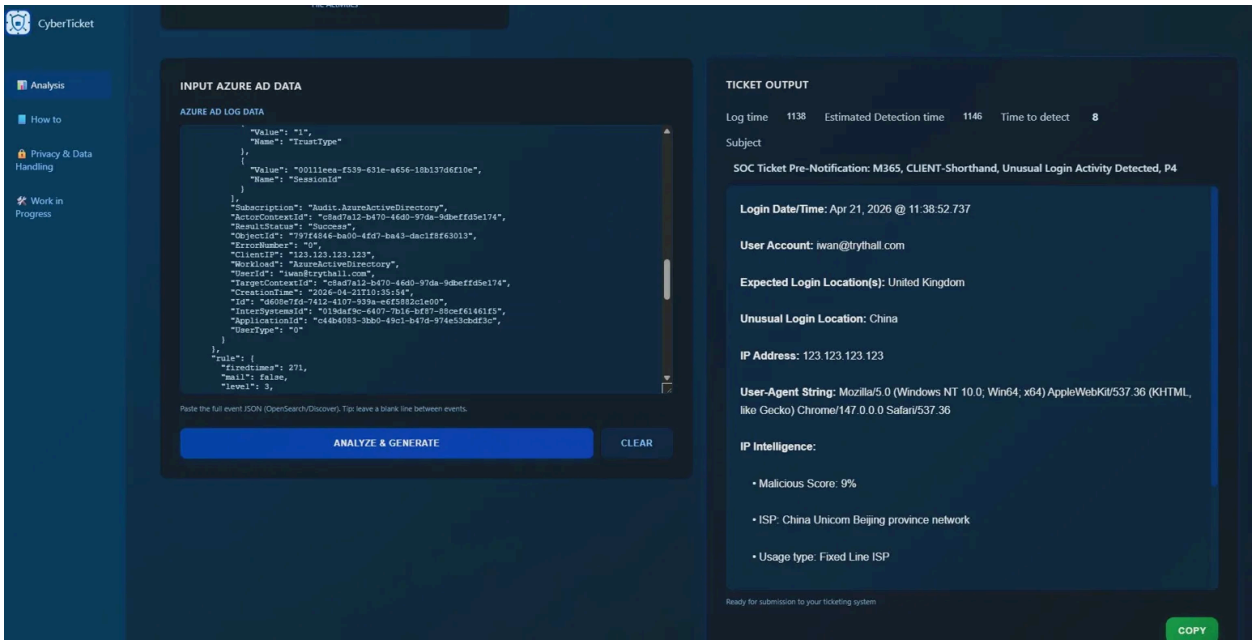


*Figure 4: Completed ticket output with time-to-detect metrics, subject line, threat intelligence, and formatted incident details ready for submission*

### Operational Efficiency Gains

| Metric | Outcome |
|---|---|
| Ticket Creation Time | Reduced from 8–12 minutes to under 2 minutes (75–85% reduction) |
| Daily Time Saved | 2–5 hours of analyst time recovered per day (20–30 alerts) |
| Ticket Standardization | 100% of M365 tickets follow a consistent, complete format |
| Threat Intel Coverage | Every ticket now includes IP reputation data automatically |

## Quality Improvements

**Reduced Human Error:** Automated field extraction eliminated transcription errors, particularly for complex fields like user-agent strings and timestamps with timezone conversions.

**Enhanced Investigative Context:** Consistent inclusion of all relevant fields (timestamps, operations, IP intelligence) provides investigators with complete information from the start, reducing the need for additional SIEM queries.

**Multi-Event Correlation:** The ability to process multiple related logs in a single ticket improved the clarity of attack narratives, particularly for account compromise scenarios involving sequential malicious activities.

## Strategic Benefits

**Performance Visibility:** The time-to-detect calculator provides quantifiable metrics for SOC maturity assessments and continuous improvement initiatives.

**Analyst Satisfaction:** Reducing tedious administrative work has allowed analysts to focus on higher-value activities like threat hunting and security research, improving job satisfaction and retention.

**Privacy Compliance:** The tool's architecture—processing all data client-side with no logging—ensures compliance with data protection requirements while maintaining full functionality.

**Cost Efficiency:** Developed internally without licensing costs, the tool demonstrates the SOC's capability to build custom solutions tailored to specific operational needs.

**Future-Proofing:** The modular design allows for easy extension to additional log types and data sources, making it adaptable to future SOC requirements beyond M365.

# 4. Conclusion

CyberTicket exemplifies how targeted automation can deliver significant operational improvements in a SOC environment. By focusing on a specific, high-volume workflow and iterating based on user feedback, the tool has achieved substantial time savings, quality improvements, and analyst satisfaction gains.

The development process demonstrates the value of empowering SOC teams to build internal tools rather than relying exclusively on commercial solutions. The privacy-first architecture, rapid iteration cycle, and zero licensing costs make CyberTicket a sustainable solution that directly addresses analyst pain points.

As the SOC evolves, the lessons learned from CyberTicket's development—particularly the importance of user-centric design and iterative refinement—will inform future automation initiatives across other aspects of security operations.

## Key Takeaways

| Area | Result |
|---|---|
| Ticket Creation Speed | 75–85% reduction in creation time |
| Format Standardization | 100% of M365 tickets fully standardized |
| Cost | Zero licensing cost with full privacy compliance |
| Process Improvement | Continuous improvement driven by analyst feedback |
| Performance Tracking | Measurable metrics via time-to-detect tracking |
| Architecture | Extensible design ready for future enhancements |

**Final Note:** CyberTicket has proven that well-designed, purpose-built tools can deliver outsized value when they directly address analyst workflows and reduce cognitive load on security teams. The tool represents a successful internal development initiative that has improved both operational efficiency and team morale within the SOC.